FACULTY OF COMPUTING

SEMESTER 2 2025/2026

SECP3213 BUSINESS INTELLIGENCE

SECTION 02

Project Phase 2

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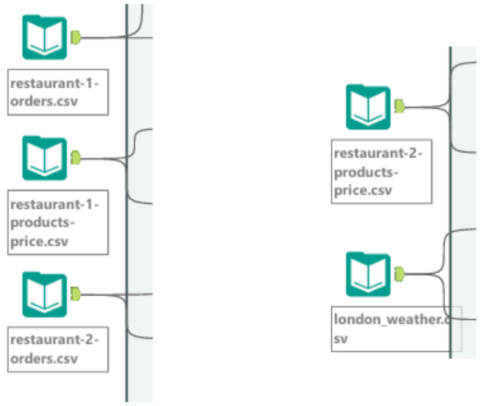
1.0 Workflow Explanation

1.1 Data Input from Multiple Datasets



Five CSV datasets were imported into Alteryx, including restaurant order data, product pricing data, and weather data. These datasets provide the information required for sales and weather-based analysis.

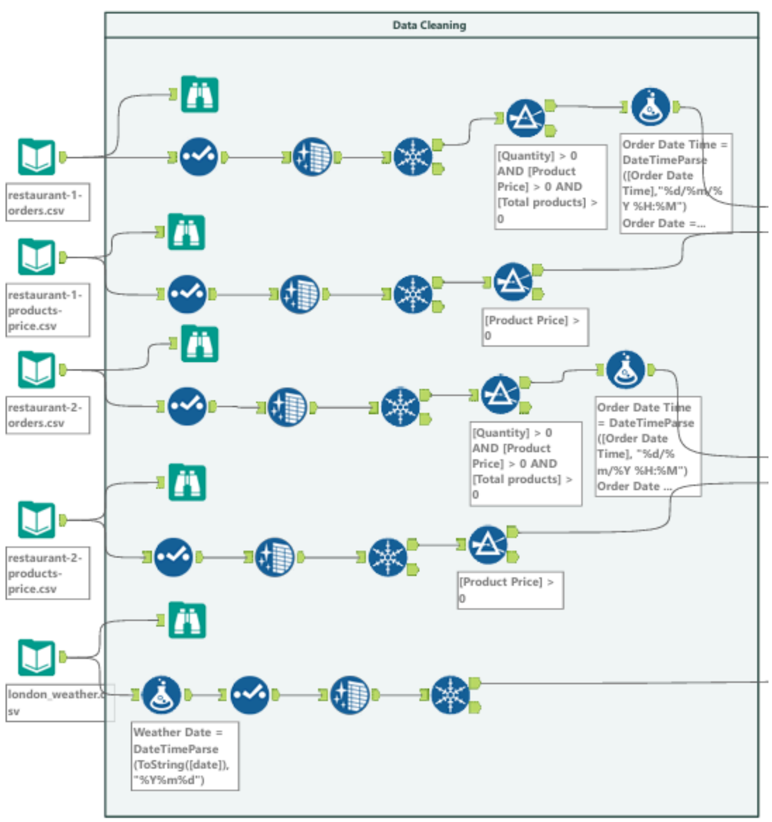
Tools & Configuration:

- **Input Data:** Configured to read the five CSV datasets and load them into Alteryx

Result Obtained:

- Multiple datasets successfully loaded.
- Data prepared for cleaning and processing.
- Foundation established for business intelligence analysis.

1.2 Data Cleaning (Missing Values and Duplication)



The Data Cleaning stage was performed to improve data quality before analysis. Multiple datasets, including restaurant orders, product prices, and weather data, were cleaned using several Alteryx tools. Filter tools were used to remove invalid records such as **Quantity > 0**, **Product Price > 0**, and **Total Products > 0**. This ensures that only valid transactions are included in the analysis. Select tools were used to verify field structures and data types, while Unique tools were used to remove duplicate records and prevent double-counting during revenue calculations.

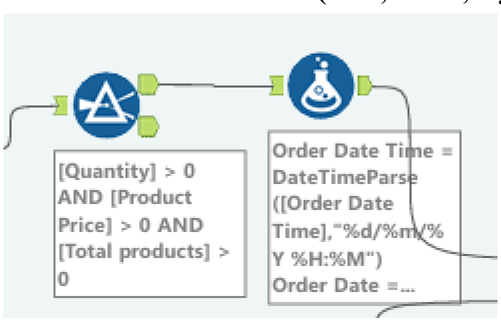
Tools & Configurations:

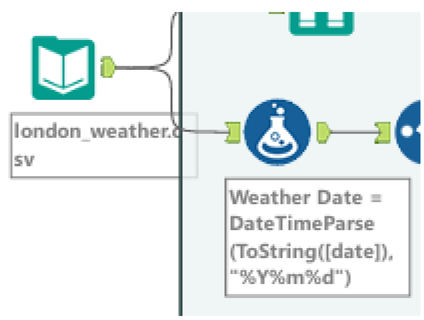
- **Browse Tool:** Used to inspect dataset contents, verify field structures and validate data quality throughout the cleaning process
- **Select Tool:** Used to standardize field data types by converting fields stored as V_String into appropriate numeric formats such as Integer and Double
- **Data Cleansing Tool:** Configured to replace null values with blanks for string fields and zero values for numeric fields
- **Unique Tool:** Used to identify and remove duplicate records

Result Obtained

- Invalid records removed.
- Duplicate records eliminated.
- Improved data quality and consistency.
- Reliable datasets prepared for further processing.

1.3 Data Transformation (Join, Filter, Aggregation)





Data transformation was performed using Formula tools to standardize date fields across the datasets. The DateTimeParse() function was used to convert **Order Date Time** and **Weather Date** into a consistent DateTime format. This ensures compatibility between datasets and supports accurate integration and time-based analysis.

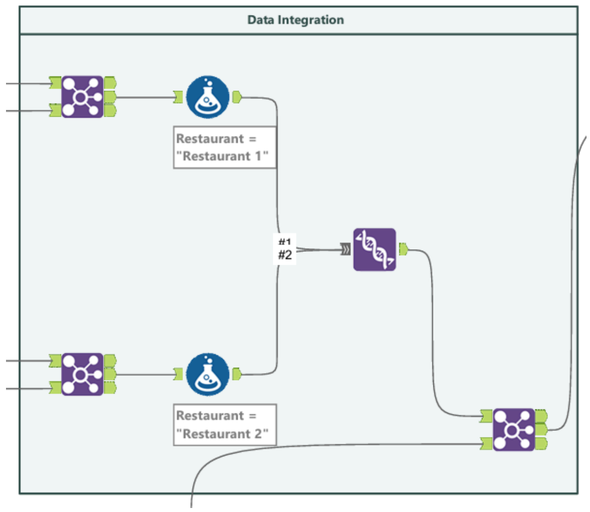
Tools & Configurations:

- **Formula Tool:** Used to standardize date fields across datasets. The weather dataset date field was converted using DateTimeParse(ToString([date]), "%Y%m%d"), while the restaurant order datasets were converted using DateTimeParse([Order Date Time], "%d/%m/%Y %H:%M")
- **Filter Tool:** Configured to retain only valid records where Quantity > 0 and Product Price > 0

Result Obtained

- Date fields standardized.
- Consistent data format achieved.
- Data prepared for integration and analysis.

1.4 Data integration (combine datasets)



After transformation, datasets from Restaurant 1 and Restaurant 2 were integrated into a single consolidated dataset. Formula tools were used to create a Restaurant identifier field with values “Restaurant 1” and “Restaurant 2”. A Union tool was then used to combine both restaurant datasets into one unified structure.

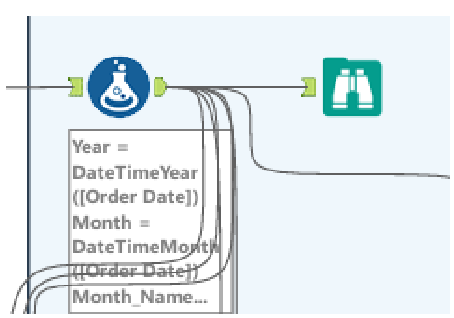
Tools & Configuration:

- **Join Tool:** Used to integrate each restaurant's order dataset with its corresponding product pricing dataset by matching the Item Name field
- **Formula Tool:** Used to create a new field named Restaurant to identify the source of each record, with values assigned as "Restaurant 1" and "Restaurant 2"
- **Union Tool:** Configured using Auto Config by Name to combine the Restaurant 1 and Restaurant 2 datasets into a single consolidated dataset
- **Join Tool:** Used again to integrate the consolidated restaurant dataset with the weather dataset using the Date field as the join key

Result Obtained

- Restaurant datasets combined.
- Unified dataset created for analysis.

1.5 Feature creation



Additional features including Year, Month, and Month_Name were derived from the Order Date field using Formula tools. These derived attributes support time-based reporting and trend analysis.

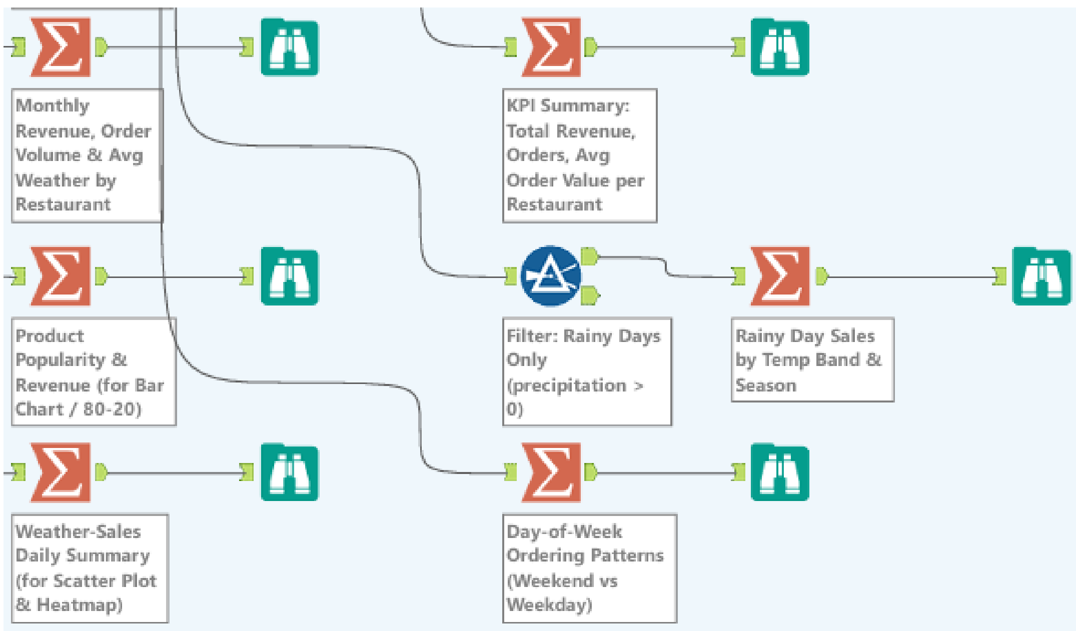
Tools & Configuration:

- **Formula Tool:** Used to extract time-based features (Month, Month_Name, Season, DayOfWeek, DayOfWeek_Num, Is_Weekend, Order_Hour, Revenue, Qty, Unit_Price, Weather_Category, Temp_Band, Cloud_Category and Year_Month) to support temporal analysis

Result Obtained

- New time-based attributes created.
- Supports trend and seasonal analysis.

1.6 Analytical Techniques



Several analyses were performed, including Monthly Revenue, Product Popularity, Weather-Sales Summary, KPI Summary, Rainy Day Sales, and Day-of-Week Ordering Patterns.

Tools & Configuration:

- **Summarize Tool:**
    1. Monthly & Seasonal Analysis: Used to aggregate monthly revenue, order volume and average weather conditions by restaurant
    2. Product Popularity Analysis: Used to identify top-selling products and revenue contribution, supporting Pareto (80/20) analysis for menu optimization
    3. Weather-Sales Summary: Used to aggregate daily sales against weather variables, enabling scatter plot and heatmap analysis of weather impact on demand
    4. KPI Summary: Used to compute key performance indicators such as Total Revenue, Average Order Value and Total Orders per Restaurant
    5. Rainy Day Sales Analysis: Used to aggregate sales specifically on rainy days

6. Day-of-Week Ordering Patterns: Used to analyze ordering behaviour by weekday vs weekend, identifying customer demand trends across the week

- **Browse Tool:** Used at multiple stages to inspect dataset structures, validate transformations and verify output correctness after each processing step
- **Filter Tool:** Configured to filter records where precipitation > 0 (rainy day)

Result Obtained:

- Sales trends identified.
- Best-selling products identified.
- Weather impact evaluated.
- Customer ordering patterns analysed.

2.0 Challenges Faced

During the development of Alteryx workflow, several challenges were encountered. One of them was ensuring consistency in data formats across multiple datasets because the weather and restaurant order datasets used different date structures. To ensure further analysis can be done successfully, DateTimeParse() function was applied to standardize all date fields into a unified format.

Another challenge was integrating multiple datasets from different restaurants with the weather dataset, which required careful use of Join and Union tools. To address this, each restaurant order dataset was first joined with its respective product pricing dataset. A restaurant identifier field was then created to label records as “Restaurant 1” and “Restaurant 2”. After that, both datasets were combined using the Union Tool to form a unified dataset. Finally, the consolidated dataset was joined with the weather dataset using the Date field as the key to enable weather-based analysis.

Lastly, another challenge was ensuring data accuracy and consistency after multiple transformation steps. This required frequent validation using the Browse Tool to check intermediate outputs and confirm that no data loss or incorrect merging occurred throughout the workflow.